

If you are new to TensorFlow, there is a very informative and easy-to-understand tutorial, 'TensorFlow for Poets' where each step is explained without any ambiguity (<https://codelabs.developers.google.com/codelabs/tensorflow-for-poets/index.html#0>).

- To add TensorFlow to the apps on Android, the following code can be added to the Gradle build:

```
allprojects {
    repositories {
        jcenter()
    }
}
dependencies {
    compile 'org.tensorflow:tensorflow-android:+'
}
```

TensorFlow has demo apps (*/tensorflow/contrib/lite/java/demo*), which are the easiest place to start. The corresponding *tflite* file can be downloaded from <https://github.com/tensorflow/tensorflow/blob/master/tensorflow/contrib/lite/g3doc/models.md> and placed inside the *assets* folder after unpacking.

The core functionality of taking an image from the preview and classifying it is given below:

```
private void classifyFrame() {
    if (classifier == null || getActivity() == null ||
        cameraDevice == null) {
        showToast("Uninitialized Classifier or invalid context.")
        return;
    }
    Bitmap bitmap = textureView.getBitmap(
        classifier.getImageSizeX(), classifier.getImageSizeY());
    String textToShow = classifier.classifyFrame(bitmap);
    bitmap.recycle();
    showToast(textToShow);
}
```

The complete tutorial provided by the official documentation can be found at <https://codelabs.developers.google.com/codelabs/tensorflow-for-poets-2-tflite/#0>.

Bender

Bender is a modern ML framework that has been built over Metal. It is built as an abstraction layer on Metal Performance Shaders (MPS). One of its important features is the ability to swiftly define and run neural networks in the iOS apps (<https://xmartlabs.github.io/Bender/#How-it-works>).

Using Bender: The official documentation describes the three basic steps required to build a ML based app.

- The model can be trained using the existing frameworks such as TensorFlow, Keras or Caffe. After this, freeze the graph files, which can then be exported.
- Import the model with Bender. The frozen graph built in the previous step can be directly imported.
- Bender can run the model on the GPU (graphics processing unit) using MPS.

More information can be gathered from <https://github.com/xmartlabs/Bender>.

To summarise, machine learning in mobile devices is extremely helpful in providing more accurate and efficient solutions to problems that require intelligence to solve them. The exponential improvements in smartphone hardware have also helped greatly in the adoption of machine learning and deep learning solutions. As these frameworks become more efficient, machine learning solutions on mobile platforms will evolve to provide better accuracy in the near future. To put it simply, machine learning combined with smartphones will bring the benefits of artificial intelligence to the masses. **END** 🐧

References

- [1] https://www.tensorflow.org/mobile/mobile_intro
- [2] <https://caffe2.ai/docs/mobile-integration.html>
- [3] <https://codelabs.developers.google.com/codelabs/tensorflow-for-poets/index.html#8>

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